A quantitative study of voiced velar nasalization in Japanese

Synopsis: This paper presents a corpus-based analysis of voiced velar nasalization (VVN) in the standard (Yamanote) dialect of Japanese. It is the first quantitative study of the phenomenon, which confirms the impressionistic observation reported in the previous literature; at the same time, our study finds that these generalizations are stochastic. Looking more closely at the determinants of this variation reveals intriguing ways in which phonological grammar and lexicon interact, as well as the role of frequency in shaping phonological variation.

Voiced Velar Nasalization: In various dialects of Japanese, [g] and [ŋ] stand in an allophonic relationship, with [g] occurring in prosodic-word-initial position, and [ŋ] occurring elsewhere (e.g. [**g**ama] “toad” vs. [ka**ŋ**ami] “mirror”). This pattern has been extensively studied in the traditional studies of Japanese (e.g. Kindaichi 1967; see also Labrune 2012) and also noted in the pre-generative European literature (Trubetskoy 1949). The relevance of this pattern to theoretical phonology was made clear by Ito & Mester (1997), which the current study substantially builds upon. The pattern of velar nasalization is reportedly sensitive to various morphophonological factors, (1)-(3):

(1) In compounds, when the second member (=N2) begins with [g] and it is a free morpheme, voiced velar nasalization targeting that [g] is optional.
 (a) E.g. N2=/ga/ “moth”, [doku-**g**a]~[doku-**ŋ**a] “poison moth”

(2) When N2 is a bound morpheme, voiced velar nasalization is obligatory.
 (b) E.g. N2=/ga/ “fang”, [doku-**ŋ**a], *[doku-**g**a] “poison fang”

(3) When N2 begins with [k], which is rendered [g] due to an independent voicing process (Rendaku), voiced velar nasalization is obligatory.
 (c) E.g. N2=/kaki/ “writing”, [joko-**ŋ**aki], *[joko-**g**ami] “horizontal writing”

(1) shows that voiced velar nasalization is not merely a static generalization about surface allophonic patterns, but it in addition manifests itself as an active (morpho)phonological alternation. Ito & Mester (1997) proposes that the optionality in (1) arises from the conflict between a force requiring VVN and output-output correspondence between the free [g]-initial form of N2 and its status in the compound. A consequence of this argument is that when N2 is bound, as in (2), output-output correspondence is impossible because there is no free [g]-initial form of N2, and hence VVN is obligatory. (3) implies a feeding process between Rendaku and VVN, and results in a saltatory phonological alternation (Hayes & White 2015): underlying /k/ is realized as [ŋ], skipping the phonetically intermediate [g], whereas underlying /g/ at least sometimes stays as [g] by output-output correspondence to a free N2. This saltatory pattern makes sense in terms of Ito and Mester’s (1997) proposal: since the free base of N2 begins with [k], not [g], output-output correspondence could not require that the bound form of N2 in compounds begin with [g].

Research aims: Our paper assesses these generalizations in a quantitative manner, and examines whether additional factors, such as lexical frequency, may affect the VVN pattern.

Methods: We extracted all compound words that had /g/-initial N2s from the *NHK Pronunciation and Accent Dictionary*, and recorded their listed pronunciations (having undergone VVN, as in [hai-**ŋ**an] "lung cancer", or not, as in [noo-**g**eka] "brain surgery"). In cases where two pronunciations were listed, we recorded the pronunciation listed as preferred or predominant. We then used the Balanced Corpus of Contemporary Written Japanese (BCCWJ) to find part-of-speech tags for N1 and N2 in each compound. We then excluded all compounds which contained either N1 or N2 with a non-noun tag, leaving 1,329 compounds with a /g/-initial N2.

Finally, we used BCCWJ to annotate each compound with the log-frequency of the compound, the log-frequency of its N1 and N2, and the number of mora in its N1 and N2. In cases where a form was absent from BCCWJ, we gave a log-frequency of 0, and annotated its mora count manually. We also extracted all compounds with /k/-initial N2s, but did not annotate them further, because they behaved categorically (see immediately below).

Results: We first assessed the evidence for the observations proposed by Ito & Mester, (1)-(3) above. Figure 1 depicts the proportion of compounds undergoing VVN, depending on the status of the N2. The vast majority of compounds with bound N2s undergo VVN (the left panel), supporting (2). Those with free N2s (the right panel) exhibit substantially more variation, supporting (1). We also found that compounds with /k/-initial N2s which undergo Rendaku categorically undergo VVN as well, supporting point (3) (not pictured here for sake of space).

Noting that there is a substantial amount of variation in VVN among compounds with free N2s (right panel of Figure 1), we demonstrate that this variation is conditioned by the relative frequencies of the compound and N2 (Figure 2), as well as the difference in frequency between the compound and N1 (Figure 3). These two findings are verified using a Bayesian logistic regression model fit using *brms* (Bürkner, 2017), with fixed effects of log compound frequency, Δlogfreq(Compound, N2), and Δlogfreq(Compound, N1).

Figure 1

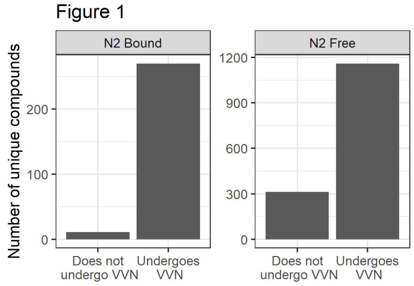


Figure 2

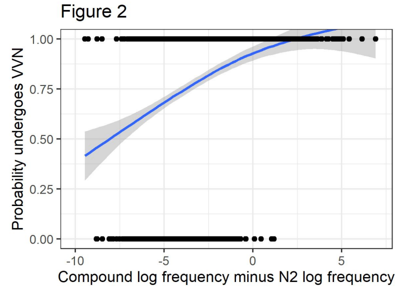
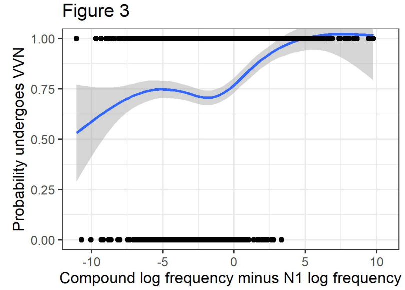


Figure 3



Discussion: Our study is the first to quantitatively verify the observations of Ito & Mester about the distribution and conditioning of [g] and [ŋ] in compounds in standard Japanese. Beyond these findings, we find that the variation noted by Ito & Mester in compounds with a free N2 is conditioned by the frequency of both N2 and N1. We argue that the effect of N2 frequency stems from the functional basis of output-output faithfulness constraints being the relative activation or accessibility of its parts: N2s which are more frequent than the compound exert more of an effect than those less frequent. The effect of N1 frequency cannot be explained the same way, however --- we speculate that as the frequency of the parts decreases relative to the compound as a whole, the compound is treated as less “compositional” and more as a monomorpheme, subject to the markedness phonotactic motivating the allophonic distribution we find in compounds with bound N2s (cf. similar findings by Rebrus & Törkenczy (2017) in the behavior of Hungarian compounds with respect to vowel harmony).

This hypothesis requires that we posit that morphological boundaries can be gradient (Hay & Baayen 2005), which can be formalized using Gradient Symbolic Representation (Smolensky & Goldrick 2016); this finding is also broadly compatible with the dual-route storage model of Zuraw (2000, 2010), Zuraw et al. (2015, 2020), or Representation Strength Theory (Moore-Cantwell, *submitted*). The current study also provides support for the thesis that the saltatory pattern is a marked phenomenon, but it can nevertheless be productive (Hayes & White 2015).